STAT 6206: Bayesian Statistics with Applications in Machine Learning Assignment 2

Academic year 25/26, second term

Due date: April 8, 2026

1. (GLMM Simulation Study): Consider clustered binary outcome data arising from $J = 20$ hospitals, with $n_j = 50$ patients per hospital. ($N = 1000$ in total). For patient i in hospital j, let

- $Y_{ij} \in \{0, 1\}$ denotes a binary outcome
- $X_{1,ij} \sim N(0, 1)$ denotes a continuous covariate
- $X_{2,ij} \sim Bernoulli(0.5)$ denotes a binary covariate
- $b_j \sim N(0, \sigma_b^2)$ denotes a random intercept, where $\sigma_b^2 = 1$

Assume the data are generated from the following logistic generalized linear mixed model (GLMM):

$$\text{logit}\{P(Y_{ij} = 1 \mid b_j)\} = \beta_0 + \beta_1 X_{1,ij} + \beta_2 X_{2,ij} + b_j,$$

with true parameter values

$$\beta_0 = -1.0, \qquad \beta_1 = 0.5, \qquad \beta_2 = 1.0.$$

Generate one simulated dataset under this model and answer the following questions.

(a) Use PyMC or other programming language to fit this GLMM model. Report the posterior mean, posterior standard deviation and 95% credible interval.

(b) Provide and interpret at least **two types of diagnostic plots** for the Bayesian fitted GLMM.

2. (Real Data Analysis): The dietox dataset is a real longitudinal dataset on pig growth, where repeated weight measurements are recorded for each pig over time.

For measurement occasion i on pig j, let

- Y_{ij} denotes the body weight (variable `Weight`),
- $X_{1,ij}$ denotes the measurement time in weeks (variable `Time`),
- $X_{2,j}$ denotes the initial body weight at the start of the study (variable `Start`),

- $b_j \sim N(0, \sigma_b^2)$ denotes a pig-specific random intercept, where the grouping variable is `Pig`.

We can write the model as following form:

$$Y_{ij} = \beta_0 + \beta_1 X_{1,ij} + \beta_2 X_{2,ij} + b_j + \epsilon_{ij},\ b_j \sim N(0, \sigma_b^2)$$

(a) Using PyMC or other programming language to fit the following two Bayesian models to the dataset:

- **Model 1:** an ordinary regression model without random effects;
- **Model 2:** a hierarchical model with pig-specific random intercepts.

For each model, report posterior mean, posterior standard deviation and 95% credible interval

(b) Compared two fitted models using **LOO**, **DIC** or other criterion. Briefly state which model performs better and explain your conclusion.